

**Notes: Chevalier and Ellison (JF, 1999)**  
**Are Some Mutual Fund Managers Better than Others? Cross-Sectional  
Patterns in Behavior and Performance**

## Contents

### **1 Big picture**

### **2 Data and measurement (key objects)**

- 2.1 Main data source and sample construction . . . . .
- 2.2 Manager characteristics . . . . .
- 2.3 Performance, risk, and style measures . . . . .
- 2.4 Other fund characteristics . . . . .
- 2.5 Descriptive facts (Table I) . . . . .

### **3 Models and empirical specifications (all equations + interpretation)**

- 3.1 Raw performance regression . . . . .
- 3.2 Risk and fund-characteristic regressions . . . . .
- 3.3 Risk-adjusted performance with fund controls . . . . .
- 3.4 Survivorship probits . . . . .
- 3.5 Heckman-style survivorship correction . . . . .
- 3.6 Management-style regressions . . . . .
- 3.7 Four-factor performance regressions . . . . .
- 3.8 Estimation choices and standard errors . . . . .

### **4 Identification and instruments (what makes the estimates credible)**

- 4.1 Why the raw correlations are not enough . . . . .
- 4.2 Systematic risk and the MBA effect . . . . .
- 4.3 Expenses, turnover, and fund-organization differences . . . . .
- 4.4 Survivorship and manager turnover . . . . .

|          |                                                                 |  |
|----------|-----------------------------------------------------------------|--|
| 4.5      | Observable investment styles . . . . .                          |  |
| 4.6      | What remains after all controls? . . . . .                      |  |
| 4.7      | What the paper can and cannot distinguish . . . . .             |  |
| 4.8      | Relation to the literature . . . . .                            |  |
| <b>5</b> | <b>Tables and figures</b>                                       |  |
| 5.1      | Table I: summary statistics . . . . .                           |  |
| 5.2      | Table II: raw performance and manager characteristics . . . . . |  |
| 5.3      | Table III: risk-taking, size, expenses, and turnover . . . . .  |  |
| 5.4      | Table IV: the main performance regressions . . . . .            |  |
| 5.5      | Table V: survivorship . . . . .                                 |  |
| 5.6      | Table VI: management styles . . . . .                           |  |
| <b>6</b> | <b>Short summary</b>                                            |  |
| <b>7</b> | <b>Conclusion</b>                                               |  |
| 7.1      | Core conclusion (what the paper establishes) . . . . .          |  |
| 7.2      | Economic intuition . . . . .                                    |  |
| 7.3      | Takeaway . . . . .                                              |  |

# 1 Big picture

- **Question.** Are some mutual fund managers better than others, and can we detect that by relating fund performance to observable manager characteristics such as age, MBA training, and the quality of the manager's undergraduate institution?
- **Setting.** The paper studies mutual funds in the United States, focusing on growth and growth-and-income funds from 1988 to 1994. The sample contains 492 managers and 2,029 fund-year observations in which a single manager had sole responsibility for the fund.
- **Core challenge.** Large raw return differences across managers need not reflect pure stock-picking ability. They can also come from differences in systematic risk, expenses, turnover, fund size, survivorship, management style, or the kinds of fund organizations that hire different managers.
- **Main conceptual contribution.** Instead of the usual "hot hands" approach that studies persistence in fund-level performance, the paper moves to the manager level and asks whether cross-sectional performance is related to manager characteristics.
- **Main characteristics studied.** The authors focus on four variables: the average SAT score at the manager's undergraduate institution, whether the manager has an MBA, the manager's age, and the manager's tenure with the fund.
- **Main raw result.** In simple excess-return regressions, managers from higher-SAT schools, managers with MBAs, and younger managers appear to outperform.
- **Main adjusted result.** After controlling for risk, expenses, turnover, fund size, survivorship, and observable investment styles, the most robust predictor of higher risk-adjusted performance is the SAT score of the manager's undergraduate institution.
- **What happens to the other raw patterns?**
  - The MBA effect disappears once systematic risk is controlled for: MBA managers mainly look better in raw returns because they manage higher-beta funds.
  - The age effect weakens materially after accounting for expenses and survivorship bias, although younger managers still look somewhat better in the four-factor specifications.
- **Practical-relevance result.** Expenses matter enormously. The paper argues that only a subset of funds are predicted to beat the market net of expenses, and that this is much more plausible for low-expense funds managed by high-SAT managers.
- **Economic takeaway.** The paper provides suggestive evidence that some managers are better than others, but it cannot cleanly separate whether the remaining performance gap reflects inherent ability, better education, better networks, stronger incentives, or simply better fund organizations.

## 2 Data and measurement (key objects)

## 2.1 Main data source and sample construction

- The main source is Morningstar. The starting point is the set of growth and growth-and-income mutual funds listed in Morningstar’s March 1994 *Mutual Funds OnDisc* CD-ROM.
- From the March 1994 CD the authors obtain monthly returns, expense ratios, assets under management, turnover ratios, and information about each fund’s current manager.
- Using additional Morningstar CDs and the *Morningstar Mutual Fund Sourcebook* at roughly annual intervals, the authors track managers backward to 1988 and forward through 1995 and check consistency using fund-name histories.
- The analysis focuses on fund-years in which Morningstar reports that one manager alone was responsible for the fund as of December 31 of the previous year. Team-managed funds are dropped because it is too hard to assign responsibility cleanly across multiple named managers.
- The main return-performance sample contains 2,029 fund-years. Some later regressions have fewer observations because assets, expenses, turnover, or school-SAT information are missing.

*Interpretation.* The paper is deliberately designed to study *managers*, not just funds. That is important because managerial turnover is high enough that manager-level and fund-level analyses can give different answers.

## 2.2 Manager characteristics

- **Manager college SAT.** For each manager, the authors assign the average composite SAT score of the undergraduate institution the manager attended, divided by 100 in the regressions.
- **MBA dummy.** Equals one if the manager has an MBA degree and zero otherwise.
- **Manager age.** Constructed approximately by assuming that each manager was 21 years old at college graduation.
- **Manager tenure.** The number of years the manager has held the current position.

The college-SAT variable is built from *Lovejoy’s College Guide*. When schools report upper and lower verbal and math SAT bounds, the authors approximate the composite SAT as

$$SAT^{school} = \frac{SAT_{low}^V + SAT_{high}^V}{2} + \frac{SAT_{low}^M + SAT_{high}^M}{2}. \quad (1)$$

When SAT data are unavailable, the appendix describes fallback procedures using SAT means, ACT ranges, ACT averages, ACT composite scores, or Lovejoy’s selectivity index.

*Interpretation.* The SAT measure is not a causal treatment. It is a proxy that may capture some mixture of ability, training quality, selectivity, and social-network value.

## 2.3 Performance, risk, and style measures

For each fund-year  $i, t$ , the paper constructs several performance objects.

### 1. Simple excess return.

$$ER_{it} = R_{it}^{fund} - R_{it}^{mkt}, \quad (2)$$

where the benchmark market return is the annual return on the value-weighted NYSE/AMEX/Nasdaq composite index.

**2. Market-model alpha and beta.** Within each year, monthly excess fund returns are regressed on monthly excess market returns:

$$R_{im}^{fund} - R_m^f = \alpha_{it} + \beta_{it}(R_m^{mkt} - R_m^f) + \varepsilon_{im}, \quad m \in t. \quad (3)$$

The estimated intercept is the annualized market-model alpha, and the slope is beta.

**3. Unsystematic risk.**

$$UnsysRisk_{it} = sd(\hat{\varepsilon}_{im}), \quad (4)$$

that is, the standard deviation of the residuals from the market-model regression.

**4. Four-factor model and style loadings.** The paper also estimates year-by-year fund loadings on the market, book-to-market, size, and momentum factors:

$$R_{im}^{fund} - R_m^f = a_{it} + b_{it}^M RMRF_m + b_{it}^H HML_m + b_{it}^S SMB_m + b_{it}^P PR1YR_m + u_{im}. \quad (5)$$

The coefficients  $b_{it}^H$ ,  $b_{it}^S$ , and  $b_{it}^P$  are the HML, SMB, and PR1YR style weights.

**5. Four-factor excess return.** Using the realized annual returns on the four factors and the estimated year-specific factor weights, the paper defines

$$Alpha4_{it} = R_{it}^{fund} - R_t^f - \hat{b}_{it}^M RMRF_t - \hat{b}_{it}^H HML_t - \hat{b}_{it}^S SMB_t - \hat{b}_{it}^P PR1YR_t. \quad (6)$$

*Interpretation.* The paper asks whether some managers beat the market simply because they load on broad factors that happened to do well, or whether they still outperform after those style exposures are removed.

## 2.4 Other fund characteristics

The analysis also uses the following controls and outcome variables:

- a dummy for growth-and-income funds (growth funds are the omitted category),
- the logarithm of assets under management,
- the expense ratio,
- the turnover ratio,
- fund age in the survivorship probits.

*Interpretation.* These variables matter because manager characteristics may proxy not just for skill, but also for the types of funds and organizations in which different managers work.

## 2.5 Descriptive facts (Table I)

- Mean simple excess return:  $-0.504\%$  per year.
- Mean market-model alpha:  $-0.502\%$  per year.

- Mean four-factor alpha:  $-0.698\%$  per year.
- Mean manager college SAT: 11.416 (that is, about 1,142 in raw SAT points).
- MBA share:  $59.6\%$ .
- Mean manager age: 44.18 years.
- Mean manager tenure: 3.79 years.
- Mean beta: 0.971.
- Mean log assets: 4.359.
- Mean expense ratio:  $1.352\%$ .
- Mean turnover ratio:  $76.81\%$ .
- Mean HML, SMB, PR1YR loadings:  $-0.020$ ,  $0.151$ , and  $0.015$ .

*Interpretation.* The average mutual fund in the sample does *not* outperform the market after expenses. That makes the cross-sectional differences especially important: even if the average fund does not add value, some types of managers might still do better than others.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Raw performance regression

The starting specification relates simple excess returns to manager characteristics:

$$ER_{it} = a + b_1 SAT_{it} + b_2 MBA_{it} + b_3 Age_{it} + b_4 Tenure_{it} + b_5 GI_{it} + u_{it}, \quad (7)$$

where  $GI_{it}$  is the growth-and-income dummy.

*Interpretation.* This is the “stylized facts” regression. It asks whether managers with certain observable characteristics appear to earn higher raw returns before correcting for risk, expenses, or selection.

### 3.2 Risk and fund-characteristic regressions

To understand whether managers systematically behave differently, the paper runs regressions of the form

$$Y_{it} = a + b_1 SAT_{it} + b_2 MBA_{it} + b_3 Age_{it} + b_4 Tenure_{it} + b_5 GI_{it} + u_{it}, \quad (8)$$

where  $Y_{it}$  is alternatively

- fund beta,
- log assets under management,
- the expense ratio,

- the turnover ratio.

When the dependent variable is expense ratio, log assets is included as an additional regressor.

*Interpretation.* These regressions show whether high-return managers simply take more market risk, work at bigger funds, charge different fees, or trade differently.

### 3.3 Risk-adjusted performance with fund controls

The main adjusted return regression replaces simple excess returns with risk-adjusted alpha and adds key fund controls:

$$\alpha_{it} = a + b_1 SAT_{it} + b_2 MBA_{it} + b_3 Age_{it} + b_4 GI_{it} + c_1 Expense_{it} + c_2 \log(Assets_{it}) + c_3 Turnover_{it} + u_{it}. \quad (9)$$

In this specification, tenure is dropped because it is not significant in the raw-return regression. The paper treats expense and turnover as endogenous and instruments for them with their lagged values.

*Interpretation.* This is the paper’s main “horse-race” specification. It asks whether the manager-characteristics effects survive after controlling for the most obvious observable fund differences.

### 3.4 Survivorship probits

To study whether the sample-selection process is correlated with manager characteristics, the paper estimates fund-survival and manager-survival probits of the form

$$Pr(Survive_{i,t+1} = 1) = \Phi(a + \lambda_1 \alpha_{it} + \lambda_2 SAT_{it} + \lambda_3 MBA_{it} + \lambda_4 Age_{it} + \lambda_5 \alpha_{it}(SAT_{it} - 10) + \lambda_6 \alpha_{it} MBA_{it} + \lambda_7 \alpha_{it} \dots) \quad (10)$$

and an analogous equation for whether the manager is still found in the dataset the next year.

*Interpretation.* These probits ask two questions:

- Are poorly performing funds more likely to disappear?
- Is that disappearance process more performance-sensitive for some types of managers than for others?

If the answer is yes, raw cross-sectional performance comparisons can be badly biased.

### 3.5 Heckman-style survivorship correction

For the pre-1993 data, the paper models selection explicitly. In simplified notation,

$$s_i = (x_{1i}\gamma)r_i + x_{2i}\delta + \varepsilon_{1i}, \quad (11)$$

$$s_i^* = \mathbf{1}\{s_i > 0\}, \quad (12)$$

$$r_i = x_{3i}\beta + \varepsilon_{2i}. \quad (13)$$

Under normality and independence assumptions, the expected return error conditional on survival is

$$E[\varepsilon_{2i} \mid s_i^* = 1, x_{1i}, x_{2i}, x_{3i}] = \frac{(x_{1i}\gamma)\sigma_i^2}{\sqrt{1 + (x_{1i}\gamma)^2\sigma_i^2}} \lambda\left(\frac{(x_{1i}\gamma)x_{3i}\beta + x_{2i}\delta}{\sqrt{1 + (x_{1i}\gamma)^2\sigma_i^2}}\right), \quad (14)$$

where  $\lambda(z) = \phi(z)/\Phi(z)$  is the inverse Mills ratio.

The paper estimates the survival equation on the reduced-survivorship-bias sample, constructs a correction term for the pre-1993 observations, and then re-estimates the return equation after subtracting that selection term.

*Interpretation.* The correction is meant to remove the part of observed pre-1993 returns that is high only because poorly performing funds or managers disappeared before they could enter the sample.

### 3.6 Management-style regressions

The authors also relate factor loadings to manager characteristics:

$$Weight_{it}^k = a + b_1 SAT_{it} + b_2 MBA_{it} + b_3 Age_{it} + b_4 Tenure_{it} + b_5 GI_{it} + b_6 \log(Assets_{it}) + u_{it}, \quad (15)$$

for  $k \in \{HML, SMB, PR1YR\}$ .

*Interpretation.* This checks whether “good” managers are simply tilted toward styles that performed well in the sample period rather than showing true within-style stock-picking skill.

### 3.7 Four-factor performance regressions

Finally, the paper repeats the main adjusted-performance regression using four-factor excess returns:

$$Alpha_{4it} = a + b_1 SAT_{it} + b_2 MBA_{it} + b_3 Age_{it} + b_4 GI_{it} + c_1 Expense_{it} + c_2 \log(Assets_{it}) + c_3 Turnover_{it} + u_{it}. \quad (16)$$

*Interpretation.* If the SAT effect disappears here, then high-SAT managers only looked good because they were in the right factor categories. If it survives, then the remaining performance difference looks more like within-category stock selection.

### 3.8 Estimation choices and standard errors

- Table II and Table IV use heteroskedasticity-robust standard errors.
- Table III and Table VI use Newey–West standard errors because the authors expect serial correlation in residuals for a given fund across years.
- The paper compares full-sample estimates, reduced-survivorship-bias estimates, and full-sample Heckman-corrected estimates.

*Interpretation.* The econometric design is not about one dramatic instrument. It is about showing, step by step, which raw effects disappear once each confound is brought into the model.

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 Why the raw correlations are not enough

The paper begins by stressing that the raw coefficients are too large to interpret as clean ability differences. For example, the simple-return regression predicts economically massive gaps by age and by school SAT. That is exactly why the rest of the paper decomposes the raw correlations into risk-taking, fee differences, survivorship, and investment style.

*Interpretation.* The identification strategy is therefore incremental: start with a striking raw pattern and then keep asking what happens when each plausible alternative explanation is stripped away.

### 4.2 Systematic risk and the MBA effect

The clearest example is the MBA result. In raw returns, MBA managers outperform non-MBA managers by about 63 basis points per year. But MBA managers also manage higher-beta funds. Once the dependent variable is changed from simple excess return to market-model alpha, the MBA coefficient collapses from about 0.63 to about 0.04.

*Interpretation.* The paper's message here is strong: the MBA result in raw returns is basically a risk-taking result, not convincing evidence of superior stock-picking ability.

### 4.3 Expenses, turnover, and fund-organization differences

Different managers also work at systematically different kinds of funds.

- Higher-SAT managers are associated with larger funds, lower expense ratios, and lower turnover.
- MBA managers also manage larger funds and somewhat lower-expense funds.
- Older managers are associated with smaller, higher-expense funds.

Because expenses and turnover may be endogenous, the paper instruments for current values using lags.

*Interpretation.* This matters because a manager can look better either because the manager is better, or because the manager works in an organization with lower costs, better support staff, or stronger incentives.

### 4.4 Survivorship and manager turnover

The paper identifies three survivorship problems:

- the 1994 Morningstar CD omits funds already dead by early 1994,
- pre-1990 manager-education data are backfilled only for managers who survive long enough to appear later,

- partial-year deaths and missing expense variables create additional sample selection.

The reduced-survivorship-bias sample and the probit estimates show that performance matters for survival, and that survival is more performance-sensitive for younger managers. In contrast, the SAT-survival interaction terms are positive but statistically insignificant.

*Interpretation.* This is why the age effect shrinks a lot once survivorship is handled, whereas the SAT effect remains much more stable.

## 4.5 Observable investment styles

The paper asks whether better-looking managers simply loaded on the kinds of stocks that happened to win in this period.

- There is little sign that higher-SAT managers systematically tilt toward HML, SMB, or PR1YR strategies.
- MBA managers tilt toward low book-to-market (“glamour”) stocks.
- Older managers show some tendency toward momentum strategies.

*Interpretation.* The SAT effect is especially interesting because it survives even though high-SAT managers do *not* appear to earn their higher returns by a simple, obvious factor tilt.

## 4.6 What remains after all controls?

The strongest residual pattern is the college-SAT effect. In the full-sample Heckman-corrected alpha regression, the SAT coefficient is about 0.291; in the full-sample Heckman-corrected four-factor regression, it is about 0.363. The age effect remains negative but is weaker once survivorship is addressed. The MBA effect is not robust in any risk-adjusted specification.

*Interpretation.* The paper’s most defensible statement is *not* that MBAs are better, and not even that young managers are definitely better. It is that managers from more selective undergraduate institutions deliver persistently higher risk-adjusted returns on average.

## 4.7 What the paper can and cannot distinguish

The paper is explicit about the limits of interpretation.

- A high-SAT school could proxy for higher inherent ability.
- It could reflect direct educational benefits.
- It could reflect better networks in finance.
- It could simply mean that such managers are hired by better fund companies with lower unreported expenses or better support systems.

Similarly, the age effect could reflect career concerns, cohort differences in education, or reverse selection if better managers leave the retail mutual fund sector as they become older.

*Interpretation.* The paper identifies *who* tends to do better, but it does not cleanly identify *why*. That is an important distinction.

## 4.8 Relation to the literature

Relative to the mutual-fund persistence literature, the paper differs in two ways:

- it studies *managers* rather than funds,
- it uses observable manager characteristics rather than persistence in each fund's own past performance.

The paper therefore complements, rather than replaces, the “hot hands” literature associated with studies such as Hendricks, Patel, and Zeckhauser, Gruber, Malkiel, and Carhart.

## 5 Tables and figures

The paper's main evidence is concentrated in Tables I–VI. Below, each table is read in the same style as your sample note.

### 5.1 Table I: summary statistics

**Read:**

- The main sample contains 2,029 fund-years.
- The average manager attended a school with SAT 11.416 (that is, about 1,142 in raw SAT points), is 44.2 years old, and has held the job for 3.8 years.
- About 59.6% of managers have MBAs.
- The mean fund beta is 0.971 and the mean expense ratio is 1.352%.
- Average simple excess return, alpha, and four-factor alpha are all slightly negative.

**Economic message.** The sample already hints that the average actively managed fund does not clearly beat the market after expenses. The empirical question is therefore about *cross-sectional differences across managers*, not about the unconditional performance of the whole industry.

**Table I**  
**Summary Statistics**

Summary statistics for all of the variables used in the analysis are presented. The observations are fund-years. The manager characteristics variables include the SAT of matriculants at the manager's undergraduate institution (divided by 100), a dummy variable that takes the value of one if the manager has an MBA degree and zero otherwise, the manager's age, and the manager's tenure. The analysis also utilizes performance and risk characteristics of each fund. Alpha is the market model excess return, in percentage per year. Beta is the coefficient of the market portfolio in a regression of the fund's monthly returns minus the risk-free rate on the monthly returns of the market portfolio minus the risk-free rate. Unsys risk is the standard deviation of the residuals from this regression, expressed in percentage points per year. The HML, SMB, and PR1YR weights are the coefficients from a regression of the fund's monthly returns on the market returns minus the risk-free rate and the returns of three other factor portfolios. The HML portfolio is a zero-investment portfolio constructed by subtracting the returns of low book-to-market ratio stocks from the returns of high book-to-market ratio stocks. The SMB portfolio is constructed by subtracting the returns of large market capitalization firms from the stock returns of small market capitalization firms. The PR1YR portfolio is the spread between the performance of stocks in the top 30 percent of returns in the prior 12 months and those in the bottom 30 percent. Alpha4 is the excess return from this four-factor model in percent per year. Other fund characteristics utilized are a dummy variable that takes the value of zero for growth funds and one for growth and income funds, the log of total fund assets under management, and the fund's expense and turnover ratios, expressed as percentages.

| Variable                   | # of Obs. | Mean   | Std. Dev. |
|----------------------------|-----------|--------|-----------|
| Simple excess return (%)   | 2029      | -0.504 | 8.451     |
| Manager college SAT (/100) | 2029      | 11.416 | 1.436     |
| Manager MBA                | 2029      | 0.596  | 0.491     |
| Manager age                | 2029      | 44.176 | 9.684     |
| Manager tenure             | 2029      | 3.793  | 5.058     |
| Growth-income dummy        | 2029      | 0.374  | 0.484     |
| Alpha                      | 2029      | -0.502 | 7.862     |
| Beta                       | 2029      | 0.971  | 0.247     |
| Log of assets              | 1907      | 4.359  | 1.913     |
| Expense ratio              | 1947      | 1.352  | 1.026     |
| Turnover ratio             | 1885      | 76.813 | 69.368    |
| HML weight                 | 2029      | -0.020 | 0.428     |
| SMB weight                 | 2029      | 0.151  | 0.389     |
| PR1YR weight               | 2029      | 0.015  | 0.228     |
| Unsys risk                 | 2029      | 2.573  | 3.402     |
| Alpha4                     | 2029      | -0.698 | 7.892     |

**Figure 1:** Table I from Chevalier and Ellison (1999).

## 5.2 Table II: raw performance and manager characteristics

**Read:**

- This is the paper's first-pass regression of simple excess returns on manager characteristics.
- The SAT coefficient is about 0.463 and is significant at the 1% level.
- The MBA coefficient is about 0.631 and is marginally significant.
- The age coefficient is about  $-0.086$ , implying that older managers look much worse in raw annual returns.
- Tenure is close to zero and insignificant.

**Economic message.** The raw data make it look as if managers from better schools, with MBAs,

and especially younger managers are much better. But the magnitudes are so large that they almost force the reader to suspect that omitted factors are driving much of the pattern.

**Table II**  
**Mutual Fund Performance and Manager Characteristics**

The dependent variable, calendar year simple excess return is regressed on a set of manager characteristics, including the average SAT score of matriculants at the manager's undergraduate institution (divided by 100), a dummy variable that takes the value of one if the manager has an MBA degree and of zero otherwise, the manager's age, and the manager's tenure with the fund. A dummy variable is also included that takes the value of one if the fund is a growth and income fund and the value of zero if the fund is a growth fund. The observations are fund-years. Heteroskedasticity robust standard errors are in parentheses.

| Independent Variables | Coefficients   |
|-----------------------|----------------|
| Constant              | -1.704 (1.756) |
| Manager college SAT   | 0.463 (0.136)  |
| Manager MBA           | 0.631 (0.391)  |
| Manager age           | -0.086 (0.022) |
| Manager tenure        | 0.005 (0.046)  |
| Growth-income dummy   | -1.836 (0.351) |
| $R^2$                 | 0.031          |
| No. of observations   | 2029           |

creases slightly with tenure, but this effect is not significantly different from zero at standard confidence levels.

Differences in stock-picking ability may be part of the explanation for the results above, although the magnitudes of the effects are so large as to suggest that other factors must be at work as well

**Figure 2:** Table II from Chevalier and Ellison (1999).

### 5.3 Table III: risk-taking, size, expenses, and turnover

**Read:**

- Managers from higher-SAT schools and managers with MBAs manage higher-beta funds.
- Higher-SAT managers also manage larger funds, lower-expense funds, and lower-turnover funds.
- Older managers are associated with smaller funds and higher expense ratios.
- Longer-tenure managers choose significantly lower betas.

**Economic message.** Table III explains why the paper cannot stop at Table II. Observable manager characteristics are correlated with important behavioral and organizational variables. Some “performance” differences are therefore really differences in risk or in the cost structure of the funds managers run.

**Table III**  
**Fund Characteristics and Manager Characteristics**

Characteristics of and actions taken by mutual funds are regressed on characteristics of the funds' managers. The manager characteristics variables include the average SAT of matriculants at the manager's undergraduate institution (divided by 100), a dummy variable that takes the value of one if the manager has an MBA degree and zero otherwise, the manager's age, and the manager's tenure. The observations are fund-years. Newey-West standard errors are in parentheses.

| Independent Variables | Dependent Variables |                     |                    |                    |
|-----------------------|---------------------|---------------------|--------------------|--------------------|
|                       | Beta                | Log of Assets       | Expense Ratio (%)  | Turnover Ratio (%) |
| Constant              | 0.788<br>(0.069)    | 4.257<br>(0.661)    | 1.911<br>(0.225)   | 143.61<br>(25.80)  |
| Manager college SAT   | 0.011<br>(0.005)    | 0.063<br>(0.054)    | -0.055<br>(0.029)  | -5.09<br>(2.19)    |
| Manager MBA           | 0.067<br>(0.016)    | 0.393<br>(0.149)    | -0.083<br>(0.054)  | -1.88<br>(5.18)    |
| Manager age           | 0.0020<br>(0.0009)  | -0.0261<br>(0.0085) | 0.0211<br>(0.0076) | -0.027<br>(0.351)  |
| Manager tenure        | -0.0055<br>(0.0020) | 0.054<br>(0.019)    | 0.023<br>(0.017)   | -0.26<br>(0.58)    |
| Growth-income dummy   | -0.131<br>(0.015)   | 0.268<br>(0.162)    | -0.006<br>(0.083)  | -14.47<br>(5.35)   |
| Log of assets         |                     |                     | -0.206<br>(0.031)  |                    |
| $R^2$                 | 0.102               | 0.036               | 0.238              | 0.020              |
| No. of observations   | 2029                | 1907                | 1895               | 1885               |

**Figure 3:** Table III from Chevalier and Ellison (1999).

## 5.4 Table IV: the main performance regressions

Read:

- Column 1 reproduces the raw simple excess-return regression: SAT positive, MBA positive, age negative.
- Column 2 switches to market-model alpha. The MBA coefficient falls from about 0.630 to about 0.042.
- Column 3 adds expenses, assets, and turnover and uses IV for expenses and turnover. The SAT coefficient falls but remains positive at about 0.306; the age effect also shrinks in magnitude.
- Column 4 repeats the specification in the reduced-survivorship-bias sample. The age effect becomes very small and insignificant, while the SAT effect remains positive.
- Column 5 applies the Heckman-style correction on the full sample. SAT stays positive at about 0.291; age stays negative at about -0.029.
- Columns 6 and 7 replace alpha with four-factor excess returns. The SAT coefficient remains strong and becomes 0.363 in the full-sample Heckman-corrected specification; the age coefficient is about -0.043 and significant.
- Expense ratios are strongly negative throughout, with coefficients around -1.5 to -2.25.

**Economic message.** Table IV is the core of the paper. It shows a pecking order of robustness:

- the MBA effect is not robust,
- the age effect is somewhat fragile and partly selection-driven,
- the college-SAT effect is the only predictor that clearly survives the full battery of adjustments.

150.203.68.71 on Wed, 08 Apr 2026 23:44:42 UTC  
All use subject to https://about.jstor.org/terms

**Table IV**  
**More Performance Regressions**

Column 1 regresses simple excess return on manager and fund characteristics; columns 2–5 regress risk-adjusted excess returns on manager and fund characteristics. In columns 3–5, instrumental variable estimation is used, treating fund expenses and turnover as endogenous variables. Column 4 displays the same regression as column 3 utilizing only the reduced survivorship bias subsample of the data. Column 5 displays the same specification again, except the full dataset is used and a Heckman-like procedure is used to correct for survivorship biases. Finally, columns 6 and 7 show the same specifications as columns 4 and 5, replacing the dependent variable with the excess returns from a four-factor model. The regressors are manager and fund characteristics. The included manager characteristics are the average SAT score of matriculants at the manager's undergraduate institution (divided by 100), a dummy variable that takes the value of one if the manager has an MBA degree and zero otherwise, and the manager's age. A dummy variable that takes the value of one for growth and income funds and zero for growth funds is also included as a regressor. Columns 3 through 7 also include the fund's log of assets, expense ratio (%), and turnover ratio (%) as regressors. In all cases, the observations are fund-years and heteroskedasticity-robust standard errors are shown in parentheses.

| Independent Variables | Excess Return Measure / Sample / Estimation Technique |                    |                   |                   |                        |                   |                         |
|-----------------------|-------------------------------------------------------|--------------------|-------------------|-------------------|------------------------|-------------------|-------------------------|
|                       | Simple/Full/<br>OLS                                   | Alpha/Full/<br>OLS | Alpha/Full/<br>IV | Alpha/RSB/<br>IV  | Alpha/Full/<br>Heck-IV | Alpha4/RSB/<br>IV | Alpha4/Full/<br>Heck-IV |
| Constant              | −1.730<br>(1.787)                                     | −0.882<br>(1.658)  | −0.237<br>(1.988) | −2.134<br>(2.401) | −1.007<br>(1.753)      | −0.540<br>(1.959) | 1.149<br>(1.492)        |
| Manager college SAT   | 0.462<br>(0.135)                                      | 0.376<br>(0.127)   | 0.306<br>(0.133)  | 0.253<br>(0.160)  | 0.291<br>(0.120)       | 0.248<br>(0.143)  | 0.363<br>(0.102)        |
| Manager MBA           | 0.630<br>(0.390)                                      | 0.042<br>(0.363)   | −0.254<br>(0.388) | 0.056<br>(0.494)  | −0.279<br>(0.346)      | 0.437<br>(0.415)  | 0.417<br>(0.295)        |
| Manager age           | −0.084<br>(0.022)                                     | −0.082<br>(0.021)  | −0.042<br>(0.021) | −0.015<br>(0.025) | −0.029<br>(0.019)      | −0.043<br>(0.020) | −0.043<br>(0.017)       |
| Growth-income dummy   | −1.835<br>(0.351)                                     | −0.766<br>(0.331)  | −0.512<br>(0.356) | 0.888<br>(0.460)  | −0.431<br>(0.336)      | 0.416<br>(0.388)  | −0.788<br>(0.288)       |
| Expense ratio         |                                                       |                    | −1.524<br>(0.325) | −1.579<br>(0.491) | −1.504<br>(0.350)      | −1.949<br>(0.298) | −2.251<br>(0.292)       |
| Log of assets         |                                                       |                    | −0.061<br>(0.124) | 0.138<br>(0.179)  | −0.018<br>(0.116)      | 0.206<br>(0.142)  | −0.267<br>(0.098)       |
| Turnover ratio        |                                                       |                    | 0.014<br>(0.005)  | 0.013<br>(0.005)  | 0.013<br>(0.004)       | −0.003<br>(0.005) | 0.003<br>(0.003)        |
| $R^2$                 | 0.03                                                  | 0.02               | 0.05              | 0.05              | 0.05                   | 0.16              | 0.10                    |
| No. of observations   | 2029                                                  | 2029               | 1705              | 872               | 1705                   | 872               | 1705                    |

884

The Journal of Finance

**Figure 4:** Table IV from Chevalier and Ellison (1999).

## 5.5 Table V: survivorship

**Read:**

- Better-performing funds are more likely to survive, as shown by the positive coefficient on alpha in the fund-survival probit.
- Better-performing managers are also more likely to remain in the dataset.
- The SAT level effect and the SAT–alpha interaction are small and statistically insignificant in both probits.
- The interaction between age and performance is negative and significant, meaning that survival is more performance-sensitive for younger managers.

**Economic message.** Table V explains why the raw youth effect is overstated: young managers who perform badly disappear more quickly. But Table V does *not* deliver the same explanation for the SAT result, which is why the SAT coefficient remains the paper's most credible residual finding.

Table V

**Manager Characteristics and Survivorship**

Two probit specifications are estimated. The first estimates the probability that a fund survives from year  $t - 1$  to year  $t$ . The second estimates the probability that a manager who is in our sample in year  $t - 1$  remains in our sample in year  $t$ . The observations are fund-years. The independent variables are managerial characteristics, performance measures, and fund characteristics. The managerial characteristics included are the average SAT score of matriculants at the manager's undergraduate institution (divided by 100), a dummy variable that takes the value of one if the manager has an MBA degree and zero otherwise, the manager's age, and the manager's tenure with the fund. Jensen's Alpha is included as the performance measure. Interactions of this performance measure and manager SAT, MBA, and age variables are also included. The fund characteristics included are the log of fund assets and the age of the fund in years. The survival process is estimated using the subset of our data from 1992–1994 for which we have more complete data on nonsurviving funds and managers. Standard errors are in parentheses.

| Independent Variables | Dependent Variables for Probit |                     |
|-----------------------|--------------------------------|---------------------|
|                       | Fund Survival                  | Manager Found       |
| Constant              | −0.065<br>(0.773)              | 0.871<br>(0.491)    |
| Alpha                 | 0.111<br>(0.058)               | 0.117<br>(0.046)    |
| Manager college SAT   | 0.069<br>(0.060)               | 0.061<br>(0.037)    |
| Manager MBA           | 0.215<br>(0.168)               | −0.217<br>(0.113)   |
| Manager age           | 0.010<br>(0.010)               | −0.0075<br>(0.0061) |
| Alpha*(SAT-10)        | 0.0055<br>(0.0076)             | 0.0037<br>(0.0056)  |
| Alpha*MBA             | 0.024<br>(0.022)               | −0.021<br>(0.017)   |
| Alpha*Mgr. age        | −0.0023<br>(0.0012)            | −0.0021<br>(0.0009) |
| Log of assets         | 0.109<br>(0.049)               | −0.017<br>(0.032)   |
| Fund age              | 0.021<br>(0.011)               | −0.0063<br>(0.0033) |
| Manager tenure        | −0.037<br>(0.020)              | 0.042<br>(0.014)    |
| No. of observations   | 942                            | 903                 |

Figure 5: Table V from Chevalier and Ellison (1999).

## 5.6 Table VI: management styles

Read:

- Higher-SAT managers do not show strong or systematic tilts toward HML, SMB, or PR1YR.
- MBA managers tilt toward low book-to-market (“glamour”) stocks, as seen in the negative HML loading.
- Older managers show some tendency to load more on the momentum factor PR1YR.
- Growth-and-income funds tilt strongly toward value (positive HML) and away from small stocks (negative SMB), which is exactly what one would expect.

**Economic message.** Table VI strengthens the SAT result. Higher-SAT managers do better, but not because they obviously ride one of the standard factors that happened to perform well during the sample period.

Table VI

**Management Styles and Manager Characteristics**

A fund-year's weightings in a four-factor model are regressed on the characteristics of fund managers. The HML, SMB, and PR1YR weights are the coefficients from a regression of the fund's monthly returns minus the risk-free rate on the market returns minus the risk-free rate and the returns of three other factor portfolios. The HML portfolio is a zero-investment portfolio constructed by subtracting the returns of low book-to-market ratio stocks from the returns of high book-to-market ratio stocks. The SMB portfolio is constructed by subtracting the returns of large market capitalization firms from the stock returns of small market capitalization firms. The PR1YR portfolio is the spread between the performance of stocks in the top 30 percent of returns in the prior 12 months and those in the bottom 30 percent. These factor weightings are regressed on fund and manager characteristics. The included manager characteristics are the average SAT score of matriculants at the manager's undergraduate institution (divided by 100), a dummy variable that takes the value of one if the manager has an MBA degree, the manager's age, and the manager's tenure with the mutual fund. The log of fund assets and a dummy variable that takes the value of one for growth and income funds and zero for growth funds are also included as regressors. Observations are fund-years. Newey-West standard errors are in parentheses.

| Independent Variables | Dependent Variables |                    |                     |
|-----------------------|---------------------|--------------------|---------------------|
|                       | HML Wgt             | SMB Wgt            | PR1YR Wgt           |
| Constant              | -0.062<br>(0.118)   | 0.288<br>(0.101)   | -0.015<br>(0.053)   |
| Manager college SAT   | -0.007<br>(-0.009)  | -0.011<br>(0.008)  | -0.001<br>(0.004)   |
| Manager MBA           | -0.053<br>(0.024)   | -0.020<br>(0.023)  | -0.013<br>(0.012)   |
| Manager age           | -0.0012<br>(0.0014) | 0.0013<br>(0.0014) | 0.0012<br>(0.0007)  |
| Manager tenure        | 0.0072<br>(0.0032)  | 0.0031<br>(0.0030) | -0.0006<br>(0.0014) |
| Growth-income dummy   | 0.158<br>(0.023)    | -0.188<br>(0.021)  | -0.013<br>(0.010)   |
| $R^2$                 | 0.044               | 0.059              | 0.004               |
| No. of observations   | 2029                | 2029               | 2029                |

Figure 6: Table VI from Chevalier and Ellison (1999).

## 6 Short summary

- The paper asks whether observable mutual fund manager characteristics predict performance.
- Using a sample of single-manager growth and growth-and-income funds, it finds very strong raw return differences by manager education and age.
- Most of the MBA advantage disappears once systematic risk is controlled for.
- Much of the age advantage of younger managers weakens once expenses and survivorship bias are addressed, although younger managers still look somewhat better in the four-factor regressions.
- The most robust result is that managers from higher-SAT undergraduate institutions earn higher risk-adjusted returns, even after controlling for risk, expenses, survivorship, and observable factor tilts.
- The paper cannot determine whether this reflects ability, education, networks, or fund-company sorting, but it does suggest that manager-level differences are real and economically meaningful.

## 7 Conclusion

### 7.1 Core conclusion (what the paper establishes)

The paper establishes that some manager characteristics are systematically related to mutual fund performance, but that most raw differences become much less impressive once the obvious confounds are handled carefully. The one characteristic that most clearly survives the paper's adjustments is the SAT score of the manager's undergraduate institution. That result remains in both one-factor and four-factor risk-adjusted regressions and is not explained away by the survivorship evidence.

### 7.2 Economic intuition

The cleanest way to read the paper is as a decomposition exercise.

- Raw returns mix together risk-taking, fees, fund quality, and stock-picking skill.
- MBA managers look better in raw returns because they hold more systematic risk.
- Younger managers look better partly because weak young managers are more likely to disappear.
- High-SAT managers still look better even after these adjustments, and they do not seem to get there through an obvious factor-loading strategy.

That is why the paper is suggestive—not definitive—evidence that some managers are genuinely better than others.

### 7.3 Takeaway

The strongest message of the paper is not that one observable credential mechanically causes performance. It is that, after controlling for a long list of alternatives, there remains a persistent cross-sectional pattern in which managers from more selective undergraduate institutions perform better on a risk-adjusted basis. That makes the paper an important bridge between the mutual-fund persistence literature and a broader human-capital view of portfolio management.